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Paleomagnetic study of thrust-sheet motion along the Rocky Mountain front in Montana

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ABSTRACT

Near Wolf Creek, Montana, the fold and thrust belt intersects the central Montana trough at the northern edge of the Helena salient. Remanent magnetizations from 18 of 31 paleomagnetic sites in the Late Cretaceous Two Medicine Formation (~79 Ma) document counterclockwise rotations of thrust-bound rocks along the margin of the salient. Results from 12 reliable sites in the upper of two thrust sheets gives a mean direction of Dec = 328.4°, Inc = 67.9°, α_{95} = 5.63°. Results from 6 reliable sites in the underlying thrust sheet yield Dec = 321.9°, Inc = 69.3°, α_{95} = 11.87°. The difference in these directions is not significant at the 95% confidence level. The thrust sheets show no relative rotation. Combining all 18 sites gives an average direction of Dec = 326.3°, Inc = 68.4°, α_{95} = 4.94°. This average direction is 25.2° ± 14.0° counterclockwise from a Late Cretaceous reference direction (Dec = 351.5°, Inc = 69.3°) established from the adjacent Adel Mountain volcanic rocks (~76 Ma). Comparison to an older (~88 Ma) pole established from the Niobrara Formation indicates 11.7° ± 11.4° of counterclockwise rotation. The paleomagnetic results, with other geologic evidence, show that both thrust sheets interacted with an oblique, lateral foreland ramp.

INTRODUCTION

Near Wolf Creek, Montana, closely spaced thrust faults carry deformed sedimentary and volcanic rocks of the Late Cretaceous Two Medicine Formation. The thrusts strike northwest-southeast and dip moderately to the southwest. To the east, these thrust faults coalesce and show left-lateral movement (Birkholz, 1967; Woodward, 1981). Local fault planes and slickensides show that left-lateral movement extends into the Wolf Creek area. Similar allochthons exist to the south in the Helena salient (Fig. 1). There, interactions with crystalline blocks of the Rocky Mountain foreland rotated the allochthons about nearby vertical axes. Eldredge and Van der Voo (1988) showed that deflection of paleomagnetic declinations records those rotations. North of the Helena salient, crystalline foreland blocks are less obvious than in southwestern Montana. Rotations of thrust sheets may, however, mark underlying lateral ramps and foreland buttresses in the thrust system. We began this paleomagnetic study to learn about the motion of thrust sheets over lateral ramps and foreland blocks.

The rocks near Wolf Creek furnish an excellent opportunity to study the relative motion of thrust sheets. Two superimposed thrust sheets coalesce in the area. Both contain nearly complete sections of the Late Cretaceous Two Medicine Formation. The stratigraphic sections, composed of a thick sequence of welded tuffs, lava flows, and volcaniclastic sedimentary rocks, are well exposed and accessible. Ample geologic evidence suggests that these thrust sheets interacted with the foreland (Bregman, 1976; Schmidt, 1972a, 1972b, 1972c, 1977a; Swenson, 1988). The nearby and contemporaneous Adel Mountain volcanic rocks (Gunderson and Sheriff, 1991) and the Fort Hays limestone (~88 Ma) in Colorado and Kansas (Acton and Gordon, 1990) provide reliable paleomagnetic directions for comparison. Thus, the Two Medicine Formation presents the opportunity to measure relative rotation between two sheets that passed over the same subsurface features.

THE TWO MEDICINE FORMATION

The Two Medicine Formation is a Late Cretaceous (Campanian) volcanic and sedimentary sequence deposited in a foreland basin. The regional environment was near shore to lagoonal (Viele and Harris, 1965) and extended laterally more than 1,000 km. Two facies of the Two Medicine Formation occur near Wolf Creek. The eastern facies consists predominantly of sedimentary rocks with nonvolcanic clasts. The western facies is thrust bounded and consists of both volcanic rocks and volcaniclastic sedimentary rocks.

The age of the Two Medicine Formation and its relationship to thrusting is unequivocal. Lyons (1944) defined the Two Medicine Formation as the basal section of the Adel Mountain volcanic field. He assigned a Late Cretaceous age based on plant fossils in a conglomerate at the base of the volcanic pile. Later, detailed mapping (Schmidt, 1972a, 1972b, 1972c, 1977a, 1977b; Schmidt and Strong, 1972) and stratigraphic correlations (Viele and Harris, 1965) showed that the Two Medicine Formation and Adel Mountain volcanic rocks are distinct units. Obradovich and Cobban (1975) determined a K-Ar isotopic age of ~79 Ma for the Two Medicine Formation.

The age of the Two Medicine Formation is similar to the nearby Elkhorn Mountain volcanic rocks (~80 Ma; Robinson and others, 1968). Two radiometric dates (81.1 ± 3.5 and 71.2 ± 2.7 Ma) from the Adel Mountain volcanic rocks (Gunderson and Sheriff, 1991) suggest that Two Medicine and Adel volcanism are also coeval. Stratigraphic relationships determined by Schmidt (1972a, 1972b, 1972c, 1977a) place the Adel Mountain volcanic field over the Two Medicine Formation. New Ar^{40}-Ar^{39} dates from the Adel Mountain volcanic rocks (Harlan and others, 1991) confirm the ~76 Ma average for the Adel Mountain volcanic rocks. The average agrees well with age and stratigraphic relationships in the area.

If the Two Medicine Formation, Elkhorn Mountain volcanic rocks, and Adel Mountain volcanic rocks were all autochthonous, they would have the same paleomagnetic direction. The Adel Mountain volcanic rocks, in the footwall beneath the Two Medicine Formation and Elkhorn Mountains volcanic rocks, are relatively undisturbed. Thus, the Adel

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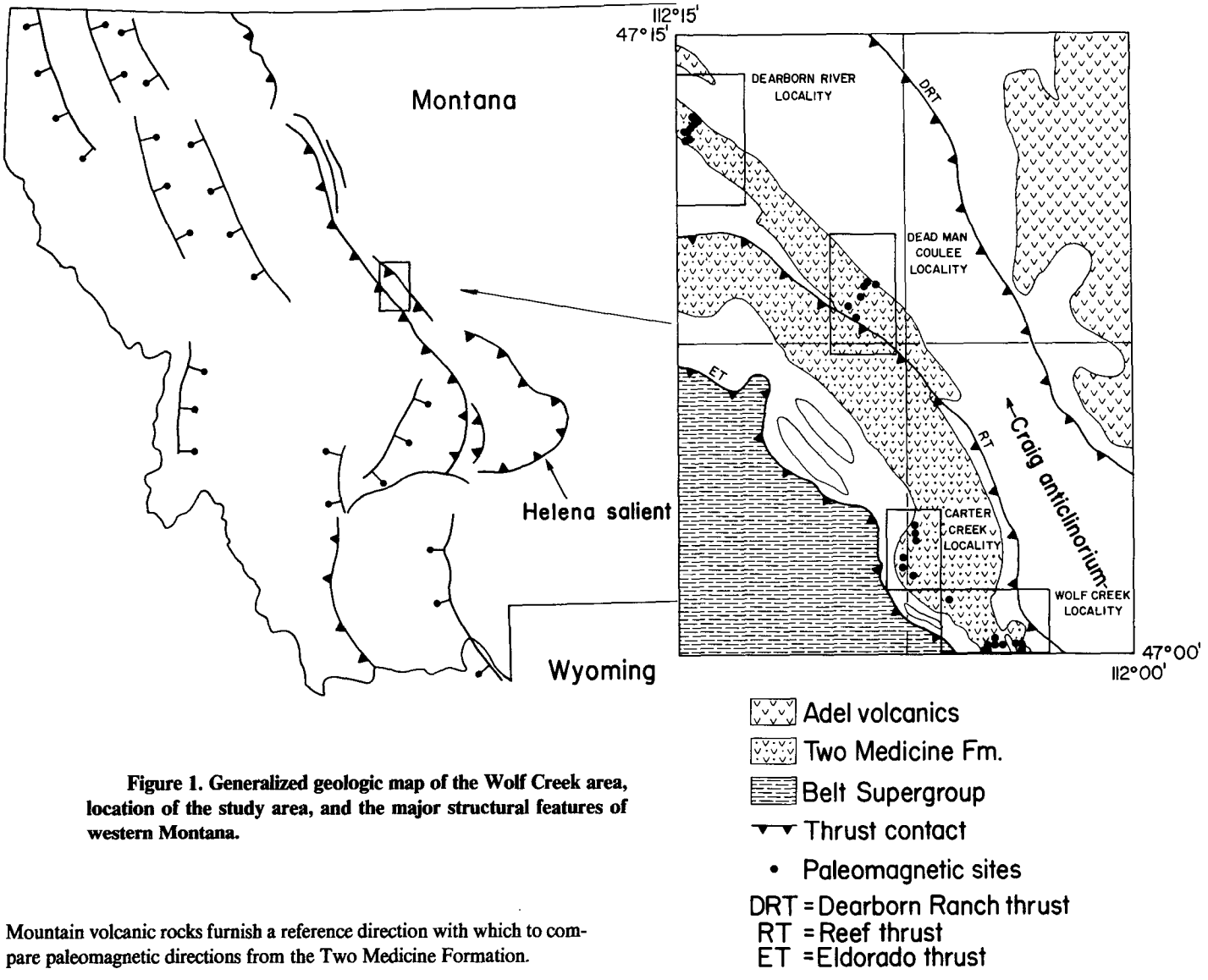


Figure 1. Generalized geologic map of the Wolf Creek area, location of the study area, and the major structural features of western Montana.

Mountain volcanic rocks furnish a reference direction with which to compare paleomagnetic directions from the Two Medicine Formation.

FIELD AND LABORATORY METHODS

To measure the relative rotation of thrust sheets, we collected paleomagnetic samples from sites in two distinct, stacked thrust sheets. To assess the paleomagnetic reliability of the rocks, we collected sites at two separate localities in each thrust sheet. This also assures a well-averaged direction for each thrust sheet. At each separate area, we sampled a long stratigraphic section to further average paleosecular variation. The rocks, located above a thrust ramp, all dip southwest or south.

This sampling scheme produced a total of 246 specimens from 31 paleomagnetic sites. The sites are all in the western facies of the Two Medicine Formation (Fig. 1). At 27 sites, we drilled samples with a standard coring device and oriented them *in situ* with a sun compass. At four sites, we collected four to five oriented hand samples. In the lab, we cored two to three specimens from each hand sample. All sites, located in prominent outcrops and roadcuts, have an average sample spacing of about 1 m.

In the lab, we employed progressive alternating field (af) and thermal demagnetization techniques to evaluate the natural remanent magnetization (NRM) of each specimen. A Molspin tumbling demagnetizing unit, capable of magnetic inductions between 2.5 and 100 mT, performed the alternating field demagnetization. Thermal demagnetization was accomplished with a homemade thermal demagnetizing unit capable of temperatures to 700 °C. Specimens remained at the demagnetizing temperature for 20 minutes and cooled in a magnetically shielded chamber with induction of less than 6 nT. We used increments of 2.5 to 25 mT for af demagnetization and 20 °C to 100 °C for thermal demagnetization. For each sample, 10 to 25 steps were measured. Both techniques were applied to at least one pilot specimen from each site. After each cleaning step, specimens were measured with a Schonstedt SSM-2A spinner magnetometer. We applied the demagnetization method that best isolated the characteristic remanent direction to the remaining specimens from a site. We also monitored magnetic susceptibility with a Bartington M.S.2 susceptibility meter to detect changes in magnetic mineralogy during thermal demagnetization.

For a few pilot specimens, af demagnetization to 100 mT did not completely separate multiple components of magnetization. For some of these specimens, more than 30% of the NRM remained after maximum

cleaning inductions. Further thermal cleaning reduced the magnetization of these specimens to a few percent of their NRM intensity. For most of these specimens, thermally isolated components paralleled the decay path established during the alternating field treatment.

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Standard orthogonal demagnetization diagrams (Zijderveld, 1967) and principal component analysis (Kirschvink, 1980) facilitated our interpretation. Of the 246 specimens collected, 184 had single component magnetizations. These had univectoral linear decays to the origin of the demagnetization diagram. Their mean directions and associated statistics were determined by the methods of Fisher (1953). For 14 of 38 samples from 5 additional sites, overlapping coercivity spectra or associated unblocking temperatures produced nonlinear demagnetization paths. To determine directions for these samples and sites, we used remagnetization circles (Halls, 1976; Kirschvink, 1980) and Bingham statistics (Onstott, 1980).

To get a site-mean direction characteristic of the original magnetic field, it is common to apply arbitrary rejection criteria at the specimen and site level. These rejection criteria are established to remove the bias introduced by outliers to small sample distributions. Our specimens had to be within two angular standard deviations of their site-mean direction. This eliminated 10 specimens. We also rejected weakly magnetized specimens ($M_0 < 10^{-4}$ A/m) and specimens with nonlinear decays, shown by maximum angular deviations greater than 7° during demagnetization (Fig. 2). This eliminated 50 specimens. Sites had to have at least 6 specimens to average and a 95% confidence limit (α_{95}) about the mean of less than 15°. To further assure independent samples of the ancient magnetic field, we combined two sites that are probably in the same trachyte flow (89TM03 and 89TM19; Table 1).

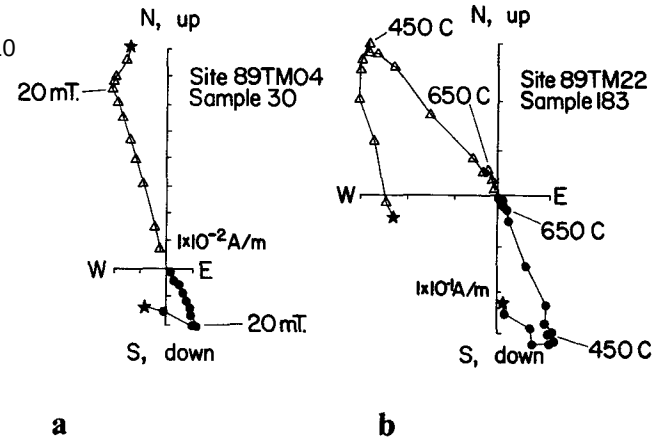


Figure 2. Representative orthogonal projections of specimens treated to stepwise demagnetization. Open symbols are for the vertical projection; closed symbols are for the horizontal plane. Stars mark the NRM positions.

Of the original 31 paleomagnetic sites, 18 passed the arbitrary rejection criteria (Table 1). The survivors include 9 sites from the Wolf Creek area, 3 from near Carter Creek, 4 from the Dead Man Coulee region, and 2 from the Dearborn River area (Fig. 1).

TABLE 1. PALEOMAGNETIC RESULTS FROM THE TWO MEDICINE FORMATION

Site	Rock	S/N	GDEC	GINC	SC	SDEC	SINC	α_{95}	k	PLAT	PLONG
					Wolf Creek locality (47.0°N, 112.0°W)						
01	WT	7/10	203.3	-63.2	170/18	163.9	-67.7	10.39	34.69	-78.8	2.9
05*	LP	4/8	352.2	40.0	155/22	332.7	42.8	10.79/18.87	-30.1/-2.1	59.0	121.8
12	LPS	8/8	206.3	-59.5	145/32	127.1	-75.2	4.25	170.77	-57.0	24.6
13	LPS	7/8	212.9	-51.3	145/32	163.2	-75.7	5.79	109.80	-71.5	43.4
14*	LPS	8/8	305.6	-39.4	170/35	341.9	-56.1	80.16	1.44	4.8	82.5
15	WT	8/8	207.8	-69.7	160/22	134.1	-75.0	5.10	118.89	-60.3	24.6
16	WT	8/8	152.9	-54.3	120/15	131.1	-60.1	2.76	403.93	-54.9	-13.8
22	LP	8/8	169.7	-47.2	155/15	153.0	-48.9	6.83	66.74	-63.0	-51.9
23	WT	7/8	197.8	-60.1	170/18	164.0	-63.6	5.32	129.72	-78.8	-25.2
29	WT	6/6	17.7	64.3	155/25	315.2	70.2	10.32/10.91	-30.8/-0.2	56.0	-174.0
31	WT	7/7	237.5	-55.8	165/35	153.2	-80.1	8.65	49.61	-63.1	48.7
					Carter Creek locality (47.0°N, 112.1°W)						
02*	SS	7/7	348.2	54.0	165/18	324.5	51.3	47.56/51.67	-2.7/-0.1	59.1	141.5
03	TP	6/8	38.6	68.8	190/15	356.8	71.0	4.35	238.39	81.4	-124.2
17	WT	7/8	326.3	77.3	245/15	330.9	62.4	6.32	92.30	69.4	159.2
18	MS	8/8	335.0	71.5	195/15	313.7	59.7	6.07	84.37	56.5	164.0
19	TP	8/8	43.2	-65.6	195/20	355.7	67.3	1.54	1295.68	85.9	-153.9
30*	CGL	8/8	141.0	-4.8	210/15	141.3	9.1	111.13	1.22	-28.1	-67.1
					Dead Man Coulee locality (47.1°N, 112.1°W)						
04	WT	7/8	202.2	-58.4	165/28	148.9	-64.3	6.81	79.62	-68.7	-13.6
06	LP	8/8	191.6	-43.4	165/28	162.0	-49.4	3.96	196.27	-68.1	-66.2
07*	LP	5/8	26.5	28.3	170/30	6.9	42.2	11.55/18.34	-22.6/-1.7	66.7	51.9
08*	SS	6/8	353.8	58.7	147/30	301.2	59.0	40.89	3.64	47.7	-170.2
09	SS	6/8	17.1	60.3	147/30	308.8	70.4	8.01	70.89	57.6	-169.6
10	SS	8/8	6.6	66.0	150/30	292.9	65.8	3.50	251.65	46.3	-175.0
					Dearborn River locality (47.2°N, 112.2°W)						
11	WT	7/8	190.8	-62.1	120/30	98.7	-80.5	7.52	65.39	-46.8	40.5
20	WT	6/7	199.3	-49.0	125/30	168.6	-75.8	7.55	79.62	-72.9	50.0
21*	SS	2/9	22.4	67.7	145/25	299.0	76.8	15.77/32.29	-10.2/-2.3	53.5	-150.9
27*	SS	5/8	62.0	23.9	140/35	70.7	57.6	51.69	3.15	39.1	-39.5

*Sites not used in final analysis.

Key: rock types (WT) welded tuff, (LP) latite porphyry, (LPS) latite porphyry stock, (SS) sandstone, (TP) trachyte porphyry, (MS) mudstone, (CGL) conglomerate; (S/N) samples used/collected; (GDEC, GINC) site mean declination and inclination; (SC) structural correction; (SDEC, SINC) site mean direction after structural correction; (α_{95}) Fisher (α_{95}) or Bingham (α_{31}/α_{32}) confidence limits; (k) concentration parameter for Fisher (k) or Bingham (k_1/k_2) distributions; (PLAT, PLONG) VGP latitude (positive north) and longitude (positive east). Four sites, too weak to measure are not included in the table.

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PALEOMAGNETIC RESULTS

Of the 18 reliable paleomagnetic sites (Wolf Creek and Carter Creek areas), 12 are in the upper thrust sheet. Six (Dead Man Coulee and Dearborn River areas) are in the lower thrust sheet (Table 1). The sites in the upper thrust sheet cover a thick stratigraphic section of Two Medicine Formation. From the bottom up, the sites are in a latite stock, two latite flows, five separate welded ash flow tuffs, a volcanic sandstone, a trachyte flow, a conglomerate, a mudstone, and a thin ash flow tuff.

The lower portion of the upper thrust sheet has reverse polarity. A viscous remanent magnetization (VRM), parallel to the present magnetic field, often obscured these primary directions. Cleaning levels of about 17.5 mT or 400 °C removes the VRM. In the Wolf Creek section, 10 of the 11 sites have 2 components of magnetization beyond the VRM (Fig. 2b). Paleomagnetic sites at the other localities (Carter Creek, Dead Man Coulee, and Dearborn River) have only a VRM and the characteristic direction (Fig. 2a). Thus, there is a local remagnetization in the Wolf Creek area.

A Late Cretaceous latite stock that cuts part of the section causes the local remagnetization in the Wolf Creek section. There are extensive quartz veins and clay alteration near the stock. The unblocking temperatures of the secondary magnetic direction show that reheating reached maximum temperatures of about 600 °C. Above 600 °C, hematite carries a characteristic magnetic direction.

The stratigraphically higher portion of the upper thrust sheet has normal polarity. These sites are in the Carter Creek area. They do not have the hydrothermal secondary magnetizations that are common in the Wolf Creek area. Rocks in the Carter Creek section include sandstone, conglomerate, a trachyte flow, an ash flow tuff, and a mudstone. Of these, the coarse sandstone and conglomerate did not provide reliable paleomagnetic results. VRM is the only secondary magnetization in the reliable sites in the normal polarity section. Cleaning to 20 mT or 570 °C removed the VRM (Fig. 2a).

The lower thrust sheet has six reliable sites from the Dead Man Coulee and Dearborn River areas (Table 1). The rocks are similar to the upper thrust sheet. Regardless, eight sites did not yield reliable paleomagnetic directions. Four sites had magnetization below the measuring capability of the magnetometer ($M_0 < 10^{-4}$ A/m). Three sites from coarse sandstones carried soft magnetizations that did not yield characteristic remanent directions. One site, in a latite porphyry, has normal and reverse directions that cannot be separated. This site yielded nonlinear decays during magnetic cleaning.

The lower thrust sheet, like the upper sheet, has a reverse polarity section overlain by rocks with normal polarity. Four of our paleomagnetic sites are in the reverse polarity section (Table 1). One of the four is in a latite flow, and the other three are in overlying welded ash flow tuffs. VRM is the only secondary magnetization in the reliable sites in the reverse polarity section. Cleaning to 15 mT or 300 °C removes the VRM.

Two reliable paleomagnetic sites (Table 1), in the Dead Man Coulee area, represent the normal polarity section of the lower thrust sheet. These sites dip about 30° and carry a prevalent secondary component of magnetization. Before tilt correction, the secondary directions group near the Late Cretaceous reference direction (Gunderson and Sheriff, 1991). Thus, the secondary directions are also near the Paleocene reference direction. Several Late Cretaceous–Paleocene dikes crosscut the thrusts in the Dead Man Coulee area. We believe that reheating by the dikes partially remagnetized these sites after motion on the thrust faults. Cleaning the specimens to 35 mT or 590 °C removed the dike's signature and reveals a characteristic remanent direction. The characteristic direction is probably a post-depositional magnetization gained during lithification.

Three tests show that the Two Medicine Formation is a reliable carrier of Late Cretaceous magnetic directions. First, the average paleomagnetic direction for the 18 reliable sites, without correcting for dip, is Dec = 14.8°, Inc = 61.8°, α_{95} = 6.04°. This is different, at the 95% confidence level, from both the present-day magnetic direction and the spin axis. When analyzed independently, the two thrust sheets give the same result. Thus, the characteristic magnetizations most likely predate the recent magnetic direction. Second, a conglomerate in the upper thrust

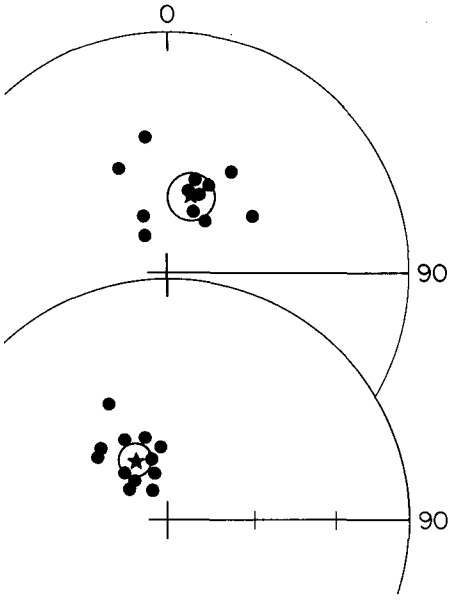


Figure 3. Equal-angle projection of 12 site mean directions from the Wolf Creek and Carter Creek areas. Reverse polarities are flipped to normal polarity. The upper diagram shows directions before correcting for local folding. The lower diagram shows directions after correcting for folding.

TABLE 2. SUMMARY OF TWO MEDICINE PALEOMAGNETIC RESULTS

	N	Dec	Inc	k	α_{95}	δ_{63}	PLAT	PLONG
Wolf Creek–Carter Creek localities in the upper thrust sheets								
Geog	12	14.7	64.4	27.19	8.48	15.54	79.9	-23.2
Strat	12	328.4	67.9	59.91	5.63	10.45	69.0	-179.3
VGP	12			28.32	8.30	15.23	68.6	174.3
Dead Man Coulee–Dearborn River localities in the lower thrust sheet								
Geog	6	14.9	56.7	79.26	7.57	9.08	75.2	14.5
Strat	6	321.9	69.3	32.78	11.87	14.15	65.2	-174.5
VGP	6			15.30	17.69	20.77	64.3	-170.6
Combined Two Medicine								
Geog	18	14.8	61.8	33.75	6.04	13.94	78.8	-6.4
Strat	18	326.3	68.4	49.91	4.94	11.46	67.8	-177.4
VGP	18			22.77	7.40	16.99	67.3	-179.8

Key: (n) number of sites used; (Dec, Inc) site-mean declination and inclination; (k) concentration parameter for Fisher distributions; (α_{95}) semiangle of the 95% confidence cone in degrees; (δ_{63}) angular standard deviation in degrees; (PLAT, PLONG) latitude (north positive) and longitude (east positive) of the paleomagnetic pole; Geog is geographic coordinates; Strat is structurally corrected coordinates. VGP pole is from tilt corrected VGP's.

sheet (Table 1) consists of poorly sorted cobbles less than 0.2 m in diameter. A 2-m-wide basaltic dike about 70 m away provides a likely source for remagnetization. We collected samples from eight different cobbles; all have well-defined characteristic directions. Yet, the directions are highly dispersed. The 95% confidence limit on their mean direction is 111°. The confidence limit would be much smaller for remagnetized cobbles. The conglomerate test confirms that the Two Medicine rocks are reliable carriers of primary remanent magnetizations. Third, a broad fold, plunging shallowly to the southwest, separates the Wolf Creek and Carter Creek localities. The well-exposed volcanic flows and sedimentary rocks provide excellent structural control. Tilt correction of the directions from these localities significantly decreases their dispersion (Fig. 3) at the 95% confidence level (McElhinny, 1964). Thus, magnetization preceded folding.

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In summary, there are 12 reliable paleomagnetic sites in the upper thrust sheet, and 6 in the lower thrust sheet (Table 1). We isolated characteristic remanent directions from several partially remagnetized sites. Post- and pre-thrusting igneous events probably caused these remagnetizations. The conglomerate test and fold test confirm the paleomagnetic stability of the rocks. The average direction of the reliable sites, before tilt correction, is not coincident with the present-day direction. Twelve acceptable sites have reverse polarity. Thus, we believe that the Two Medicine Formation is a reliable carrier of primary magnetic directions.

Table 2 summarizes the paleomagnetic results for the two thrust sheets. Before tilt correction, the upper thrust sheet has a mean paleomagnetic direction of Dec = 14.7°, Inc = 64.4°, α_{95} = 8.48° from 12 reliable sites. The lower thrust sheet has Dec = 14.9°, Inc = 56.7°, α_{95} = 7.57° from 6 reliable sites. After tilt correction, the upper thrust sheet has Dec = 328.4°, Inc = 67.9°, α_{95} = 5.63°. The lower thrust sheet has Dec = 321.9°, Inc = 69.3°, α_{95} = 11.87°.

RELATIVE ROTATIONS OF THE TWO THRUST SHEETS

To test the hypothesis that significant rotations occurred between the two thrust sheets, we compared their mean declinations. In both thrust sheets, our samples represent large, approximately time-equivalent stratigraphic sections of the Two Medicine Formation. We have 12 sites in the upper thrust sheet and 6 in the lower. Several of the latter are in sedimentary rocks. Regardless, in either thrust sheet, we may have too few independent sites to average paleosecular variation sufficiently. Given this qualification and using α_{95} as the test statistic, we calculated the relative rotation between the thrust sheets. The average paleomagnetic declination is 321.9° for the lower thrust sheet and 328.4° for the upper thrust sheet. By the method of Beck (1980) and Demarest (1983), a statistically insignificant rotation of 6.5° ± 30.2° occurred at the 95% confidence level. The data thus indicate that little or no rotation occurred between the two thrust sheets.

Next, we tested the hypothesis that the two thrust sheets rotated with respect to autochthonous rocks on the craton. The two thrust sheets are not statistically different at the 95% confidence level. Thus, we combined them for comparison to the Late Cretaceous reference direction. The combined, structurally corrected average is Dec = 326.3°, Inc = 68.4°, α_{95} = 4.94° for 18 reliable paleomagnetic sites. The paleomagnetic pole derived from the 18 virtual geomagnetic poles (VGP's) is at 67.3°N, 179.8°W, α_{95} = 7.40° (Fig. 4).

For a valid test, the 18 VGP's from the Two Medicine Formation must accurately represent the paleomagnetic pole position. Paleosecular variation must be averaged to a constant value; this is often difficult. For example, Diehl and others (1988) collected 239 mid-Tertiary VGP's from 16 studies. Yet their normal and reverse subsets yield significantly different paleomagnetic poles. In a separate study, Calderone and others (1990) got only 64 independent VGP's from 190 middle Miocene lava flows in California and Arizona. Limited subsets from these studies would not always yield an accurate representation of the true pole.

Several lines of evidence suggest that the combined Two Medicine data may adequately average paleosecular variation. First, there are both normal and reverse polarities from a distinct stratigraphic sequence. The sets are small, but the normal and reverse subsets are antipodal at the 95% confidence interval. Second, the Two Medicine section at Wolf Creek is about 1,300 m thick. Sedimentary rocks make up half of this section. Using a sedimentation rate of 7.8 cm/1,000 yr (Wallace and others, 1990), the Two Medicine sediments were deposited in about 8.3 m.y. If this estimate is accurate within an order of magnitude, enough time elapsed to average paleosecular variation. Third, paleosecular variation model G (McFadden and others, 1988) predicts 18.2° of VGP scatter at Wolf Creek's paleolatitude. The observed angular dispersion around the Two Medicine paleopole is 17.0°, with lower and upper 95% confidence intervals of 14.2° and 22.1° (Cox, 1969). Thus, at 95% confidence, our observed angular dispersion matches the predicted angular dispersion. We believe that it is proper to compare the combined Two Medicine pole to the Late Cretaceous reference pole.

The position of the Late Cretaceous reference pole is equivocal. Between the mid-Cretaceous and Paleocene, there is a 13° gap in North America's apparent polar wander path. This gap hinders the understanding of local-scale to lithosphere-scale structure developed during the Late Cretaceous. Several paleomagnetic studies provide paleomagnetic data for this gap (for example, Vugteveen, 1981; Swenson and McWilliams, 1988; Hanna, 1967, 1973; Diehl, 1992). Gunderson and Sheriff (1991) discussed possible problems with using these paleomagnetic studies of Late Cretaceous rocks, however. For example, the studies are all from allochthonous structural assemblages.

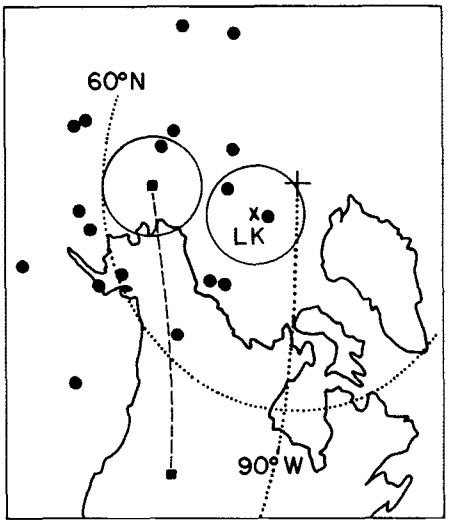


Figure 4. Equal-angle projection of the 18 reliable virtual geomagnetic poles. The dashed line connects their mean, and its 95% confidence limit, with the sampling area. LK is the Late Cretaceous pole position from the Adel Mountain volcanic field with its 95% confidence limit. The plus marks the spin axis. The land masses are North America, Baffin Island, and Greenland.

Two paleomagnetic studies of Late Cretaceous rocks from stable locations on North America are available. Acton and Gordon (1990) gave a paleopole from the Fort Hays limestone (~88 Ma) in Colorado and Kansas. Their pole position is 75°N, 185°E, α_{95} = 3.0°. This is displaced slightly toward the Paleocene pole (Diehl and others,1983) from the preferred mid-Cretaceous pole (Globerman and Irving, 1988). The Fort Hays pole records the onset of apparent polar wander at 88 Ma that followed a long interval of little or no apparent polar wander.

Gunderson and Sheriff (1991) provided a Late Cretaceous paleopole (~76 Ma) from the Adel Mountain volcanic rocks. These mildly folded and faulted rocks lie largely east of the leading edge of the fold and thrust belt. The paleomagnetic directions pass the fold test. Yet, all 26 sites have normal polarity and one expects some reversals in Late Cretaceous rocks. The paleomagnetic pole calculated from 26 VGP's from the Adel Mountain volcanic rocks is at 82.0°N, 209.9°E (α_{95} = 6.80°). This paleomagnetic pole is not distinguishable from the Paleocene reference pole (Diehl and others, 1983). It is, however, displaced slightly toward the preferred mid-Cretaceous pole (Globerman and Irving, 1988).

Comparing the pole positions (Acton and Gordon, 1990; Gunderson and Sheriff, 1991) shows 8° of apparent polar wander between deposition of the Fort Hays limestone (~88 Ma) and eruption of the Adel Mountain volcanic rocks (~76 Ma). Comparing our results (Dec = 326.3°) from the Two Medicine Formation with these poles yields different estimates of rotation of the thrust sheets. In our study area, the expected declination from the Fort Hays pole is 338.0°. This indicates a counterclockwise rotation of 11.7° ± 11.4° for the Two Medicine rocks. The expected declination from the Adel Mountain pole is 351.5°. This indicates a counterclockwise rotation of 25.2° ± 14.0°. Thus the minimum counterclockwise rotation significant at the 95% confidence level is 0.3° for the Fort Hays pole or 11.2° for the Adel Mountain pole.

Choosing the proper pole for comparison, and therefore the most likely rotation, is difficult. Only random paleomagnetic errors are included in the 95% confidence limits; normal geologic errors are not. The true uncertainty in the rotation is larger than the calculated value. Given additional uncertainty, the minimum 0.3° of significant rotation at the 95% confidence level with respect to the Fort Hays pole could as well represent some clockwise rotation. The maximum calculated significant rotation (95% confidence) of 39.2° is relative to the pole from the Adel Mountain volcanic rocks. This is not a maximum end member because the Adel Mountain volcanic rocks may be carried on a blind thrust. Although they probably experienced minimal displacement, it is possible that the Adel Mountain volcanic rocks rotated in the same direction as the Two Medicine Formation.

We are inclined to use the pole from the Adel Mountain volcanic field for comparison to our results from the Two Medicine Formation. This indicates 25.2° ± 14.0° of counterclockwise rotation around a nearby vertical axis. Although there are uncertainties associated with the ages of the rocks, the Two Medicine Formation (~79 Ma) is a bit closer in age to the Adel Mountain volcanic rocks (~76 Ma) than to the Fort Hays Limestone (~88 Ma). Of course, one could choose either pole for comparison or use a point in between. For example, increasing the observed inclination of the Fort Hays direction to allow for flattening during lithification would make the calculated rotation closer to that calculated from the Adel Mountain pole. Most choices would show that, as the thrust sheets moved along the northern margin of the Helena salient, they experienced significant counterclockwise rotation at the 95% confidence level.

DISCUSSION AND CONCLUSIONS

The overthrust belt of the northern Rocky Mountains has a sinuous trace from Utah through Alberta. Thrust sheets are locally rotated along the margins of salients in the overthrust belt. Grubbs and Van der Voo (1976) and Schwartz and Van der Voo (1984) document rotations in the Wyoming salient. They attribute rotations as large as 60° to buttressing against the ancestral Teton and Gros Ventre mountain ranges. Eldredge and Van der Voo (1988) describe rotations of as much as 54° along the southern margin of the Helena salient in southwestern Montana. These rotations may result from buttressing against the Rocky Mountain foreland. Using the Adel Mountain pole, our results indicate that two thrust sheets carrying Two Medicine Formation rotated at least 11.2° counterclockwise at the northernmost edge of the Helena salient. Their maximum rotation at the 95% confidence level is 39.2°. This rotation probably was also due to interaction with underlying crystalline rocks. If so, the effects of buttressing are local, and drag and rotation would not be transmitted far into the thrust sheets.

There are similarities between the rotation of our Two Medicine sites and equivalently thrusted rocks 130 km to the south at the southern edge of the Helena salient (Fig. 1). There, the rocks rotated clockwise (Eldredge and Van der Voo, 1988) in response to right-lateral slip on the thrust faults. The rotation near the south edge of the Helena salient mirrors that at the north edge. Near Wolf Creek, left-lateral movement on thrust faults is coincident with counterclockwise rotation.

Despite similarities between the northern and southern margins of the Helena salient, there is a significant geologic difference between the two. Large-scale uplifts near the southern margin juxtapose thrust-bound sedimentary and volcanic rocks against older, highly fragmented crystalline rocks (Schmidt and others, 1988). In contrast, foreland uplifts do not crop out in the Wolf Creek area near the north edge of the Helena salient. There, the foreland ramp is probably more uniform and has less dip than near the southern margin. Regardless, there is abundant evidence for strong interaction between thrusting and the foreland.

The Craig anticlinorium (Fig. 1) is one example of a structure most likely produced by interaction between thrusting and the foreland near Wolf Creek. It is a northeast-verging, crescent-shaped structure with secondary doubly plunging structures that have overturned northeast limbs. The change in structural style of the overthrust belt near Wolf Creek is also evidence for interaction between the foreland and the thrust sheets. Northwest of Wolf Creek, there are several imbricate thrust sheets that dip moderately to the southwest. The thrusts in the imbricate stack coalesce southeast of Wolf Creek and form a single thrust with shallow dip and sinuous trace. Bregman (1976) suggested that buttressing produced this change in tectonic style.

A long-lived basin beneath the Adel Mountain volcanic field provides evidence for topography on the buried foreland. The Missouri River follows this northeast-trending structural depression (Lyons, 1944). The contact of Adel Mountain volcanic rocks and the underlying Montana Group sedimentary rocks (Swenson, 1988) shows the trough is more than 1,000 m deep. Swenson's (1988) mathematical model suggests that the structural depression is only partially explained by elastic subsidence caused by loading. Therefore, the structure existed before the Adel Mountain volcanic field that provides the load, and it might have influenced overlying thrust sheets.

The thrust sheets in the Wolf Creek area interact with the foreland,

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despite the lack of outcropping, large-scale, crystalline uplifts common in southwest Montana. Rotations at the south edge of the Helena salient probably result from drag on an obliquely oriented foreland ramp (Eldredge and Van der Voo, 1988). The same mechanism explains the tectonic features of the Wolf Creek area. Thrust-bounded rocks of the Two Medicine Formation encountered rocks of the foreland as the thrusts progressed eastward. The oblique orientation of the foreland ramp caused the thrust sheets to rotate as they climbed it.

The hypothesis that the allochthons moved up an oblique ramp explains the tectonic features of the region and the observed rotations. The upper and lower thrust sheets moved together. They show no differential rotation. This indicates that thrust faults advanced from west to east (Hoffman and others, 1976). The upper thrust sheet was riding piggyback when the thrust sheets encountered the footwall ramp. Coupling with the footwall caused the left-lateral movement observed on the margin. Several imbricate thrust faults in a re-entrant to the north (Bregman, 1976) gather into their roof fault near Wolf Creek. Re-entrants in this area often formed over duplexes (Dohlberg, 1986) in the ductile Marias River Shale. The oblique ramp most likely impeded the southward development of a duplex and caused the thrust faults to coalesce.

The overthrust belt maintains a northwest strike for 100 km north of Wolf Creek. There, near Haystack Butte, the overthrust belt turns north. The northward extent of rotations in the overthrust belt is unknown. The change in strike probably occurs where thrust fault transport becomes perpendicular to the underlying ramp. In this case, the rotations we measure in the Wolf Creek area should not extend beyond Haystack Butte. We expect that as we work north, the rotations will decline at Haystack Butte. Toward the center of salients, the lateral extent of rotations should get smaller or disappear. Rotations will only be associated with local footwall ramps. One exception exists. On the northwest margin of the Helena salient, and east of the Idaho Batholith, the crustal-scale Purcell Anticlinorium plunges out along the Lewis and Clark fault zone (Sears, 1988). There, we anticipate large rotations of *en echelon* anticlines on the crustal ramp.

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